Session 2 (struggle day)

Putnam 2017 A1 · up to 2.5 hours

How this session works

This is a **struggle day**. No new theory. One problem, up to 90 minutes of real effort, then a careful write-up, then reflection. It uses only what Session 1 taught: small cases, contradiction, closure arguments in the spirit of induction, and the write-up discipline.

Before you start (10 min). From memory, without opening Session 1:

- Explain Cantor's diagonal argument out loud using the 3-string game.
- State the two-base-case trap in induction.
- Re-prove that there are infinitely many primes of the form $4k - 1$, on paper, in full.

Then check against Session 1. If any of the three is shaky, fix it now; the struggle problem can wait ten minutes.

Rules for the struggle block.

1. Timer on. Write the start time at the top of your paper.
2. Keep a running "ideas tried" list on the page, including the ones that failed. Failed ideas are data, not waste.
3. Hints are on the last pages, staged. No Hint 1 before 30 minutes, no Hint 2 before 50, no Hint 3 before 75.
4. Solutions are in `session02_solutions.pdf`. Do not open it until your write-up is done, even if you think you solved it.
5. If you solve it, you are *not* done. The write-up is the point.

What to hand back. Fill in `session02_writeup.tex` with your written solution, the ideas-tried list, and the reflection. That file is what gets graded.

1 The problem — up to 90 min

Problem 1 (Putnam 2017, A1)**.** Let S be the smallest set of positive integers such that

(a) 2 is in S,

(b) n is in S whenever n^2 is in S, and

(c) $(n+5)^2$ is in S whenever n is in S.

Which positive integers are not in S?
(The set S is "smallest" in the sense that S is contained in any other set of positive integers satisfying (a), (b), (c).)

Why this problem. It is a genuine A1. It is solved with exactly the tools from Session 1: compute small cases, prove a closure property by the same reasoning as an induction step, and give a two-direction argument. And the write-up has a classic trap: the word "smallest" is a hypothesis, and most failed solutions never use it.

Checkpoints. Not hints, just honest questions.

- **15 min:** have you computed at least ten elements of S by hand?
- **30 min:** do you have a *conjecture* for the answer? Write it as a full sentence before doing anything else.
- **45 min:** a question of the form "which integers are *not* in S" needs *two* proofs: one that certain integers are in S, and one that the others are not. Have you identified both?

2 Write-up 30 min

Write the main problem up as if to a Putnam grader, in `session02_writeup.tex`. Before writing, run this checklist, which is the guide's Section 1.5 condensed:

- ☐ **State the answer or claim in the first line.** Do not bury the lead.
- ☐ **Every hypothesis used?** For the main problem: (a), (b), (c), *and* "smallest." An unused hypothesis almost always means a hole.
- ☐ **No claim you know to be false.** A grader who sees one stops reading.
- ☐ **Messy? Restructure first.** Pull out a lemma. A clean lemma plus a three-line main argument beats a page of cases.
- ☐ **Name the contradiction.** If you argue by contradiction, the last sentence says exactly what impossible thing you reached.

If you solved the main problem in under 60 minutes and have time left:

Problem 2 (Putnam 2005, A1)**.** Show that every positive integer is a sum of one or more numbers of the form $2^r 3^s$, where r and s are nonnegative integers and no summand divides another. (For example, $23 = 9 + 8 + 6$.)

This one is strong induction, with the case split and the choice of basis being the whole problem.

3 Reflection and close-out 15 min

1. *Now* open `session02_solutions.pdf`. Compare line by line against your write-up. For each difference decide: cosmetic, or a real gap?
2. In the write-up file, answer: minutes to first correct conjecture on the main problem; minutes to a proof you believed; which hint, if any, you used; the idea that unlocked it, in one sentence; one thing in your write-up you would lose points for.
3. Log the session in `logs/ch01_communication_tools.md`: fill the rows for Exercises 1.49 and 1.33, add a line to the running log.

4. In 2 or 3 days, re-derive the main problem from a blank page without this sheet. That re-derivation is where the skill locks in. Put it on your calendar now.

Hints for the main problem

Staged at 30, 50, and 75 minutes. Scroll only when the timer says so.

Hint (1, after 30 min)**.** Compute concretely. $2 \in S$, so $(2+5)^2 = 49 \in S$, so $7 \in S$. Keep going. What residues mod 5 appear? Which never can?

Hint (2, after 50 min)**.** Two directions. (i) Find a set T that obviously satisfies (a), (b), (c); then "smallest" gives $S \subseteq T$. (ii) Show $T \subseteq S$ by actually generating every element of T from 2 using the rules. For (ii), first prove a small lemma: if $n \in S$ then $n + 5 \in S$. Why?

Hint (3, after 75 min)**.** The $n + 5$ lemma means you only need *one* element of S in each residue class mod 5 that occurs, and you need it to be the smallest possible one. You already have 2 and 7. Chase: $49 \to 64 = 49 + 15 \to 8$. Then find a square that is $\equiv 1 \pmod 5$ and provably in S, take its root, and keep walking down. Target: show $2, 3, 4, 6 \in S$.

Hint for the optional problem (Putnam 2005 A1)

Hint. Strong induction on n. Even n is easy from $n/2$. For odd n, subtract the largest power of 3 that fits and look at what is left. Why can none of the leftover summands divide, or be divided by, that power of 3?